A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Soils: fertility, erosion & management - Set 2

Mark scheme

AQA spec 3.2.5; The physical environment; textbook pp. 158-169

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** B, C and D – each loses soil faster than the 1 t ha^{-1} yr^{-1} at which it forms, so the soil is getting shallower **1** Accelerated erosion means the natural equilibrium between soil formation and erosion has been broken – field A loses only 0.8 t ha^{-1} yr^{-1}, which is less than the rate of formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (24.0 - 9.0) / 24.0 x 100 **1** = 62.5% (accept 62 - 63%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **contour ploughing** → ploughing along the slope rather than up and down it **Do not credit** • A bare wrong answer with no working.	2
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 Contour ploughing:** furrows run along the contour at 90° to the slope so runoff is stopped by the furrow and loses kinetic energy – the particles it was carrying are deposited in the furrow instead of being washed away **1 Terracing:** the slope is cut into a series of narrow level fields held by retaining walls, so water slows as it crosses each field and much of it infiltrates rather than running off **1 Rows of stones:** stones laid along the contours of a gentle slope slow the runoff and trap disturbed soil behind them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Contour ploughing** → ploughing along the slope rather than up and down it • **Terracing** → cutting the slope into level steps	3

Do not write outside the box

Question	Marking guidance	Marks
01.4	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1 Multicropping:** growing more than one crop in the field at a time, sown and harvested at different times, so some crop always covers the soil **1 Strip cropping:** wide strips of different crops, most effective at 90° to the prevailing wind, for farms that use machinery **1** Growing a long-term crop such as permanent grassland or a bush or tree crop, which does not need frequent replanting or soil disturbance **1** No-tillage cultivation / direct drilling of seed, which disturbs the soil far less than ploughing and sowing **1** Adding organic matter such as mulch or manure – surface mulch shields the soil from the wind and from rain splash, and the humus formed as it decomposes makes the particles adhere to each other **1 Controlling livestock:** lower stocking density and restricted movement so there is less trampling damage and more plant cover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
01.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Soil particles carried into rivers are deposited where the water slows, so the flow capacity of the river falls and it overflows more readily after heavy rain **1** Sediment fills reservoirs, reducing the volume of water they can store and shortening their useful life **1** High turbidity in the river reduces light penetration and smothers the river bed, damaging aquatic life **1** Wind erosion raises the level of atmospheric particulates, which worsens respiratory illness such as asthma **1 Erosion contributes to desertification:** vegetation struggles to grow on the thinner soil, which exposes more soil and reduces rainfall further **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **particulates** → fine solid particles / smoke and dust **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-166; AQA spec 3.2.5.2.	

Do not write outside the box